Student Handout

Name: _____ **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Name: _____ **Date:** _____ **Class:** _____

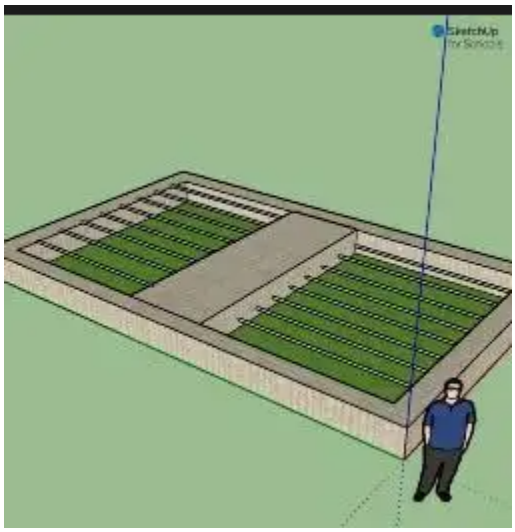
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Sammi's design

Date: 2/19/25
Description: My irrigation system is a drip irrigation system. The plants are in the box which makes the system portable. The pvc pipes running through the top of the box has the water running through drip on the plants. The box also has abilities to connect to an outside source that holds the water.

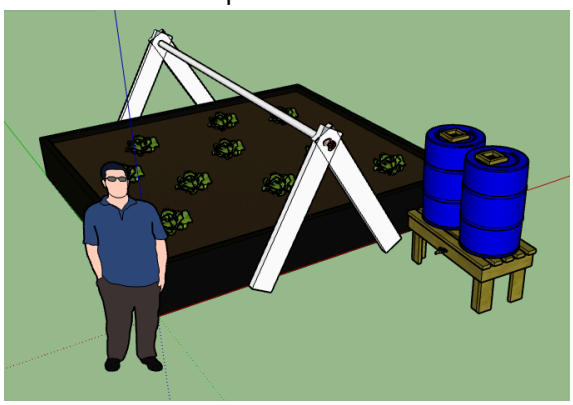


Peter's Design

Date: 2/24/25
Description: My design is 2 2x4 stands holding a PVC pipe with small holes in it. In which the

Name: _____ **Date:** _____ **Class:** _____

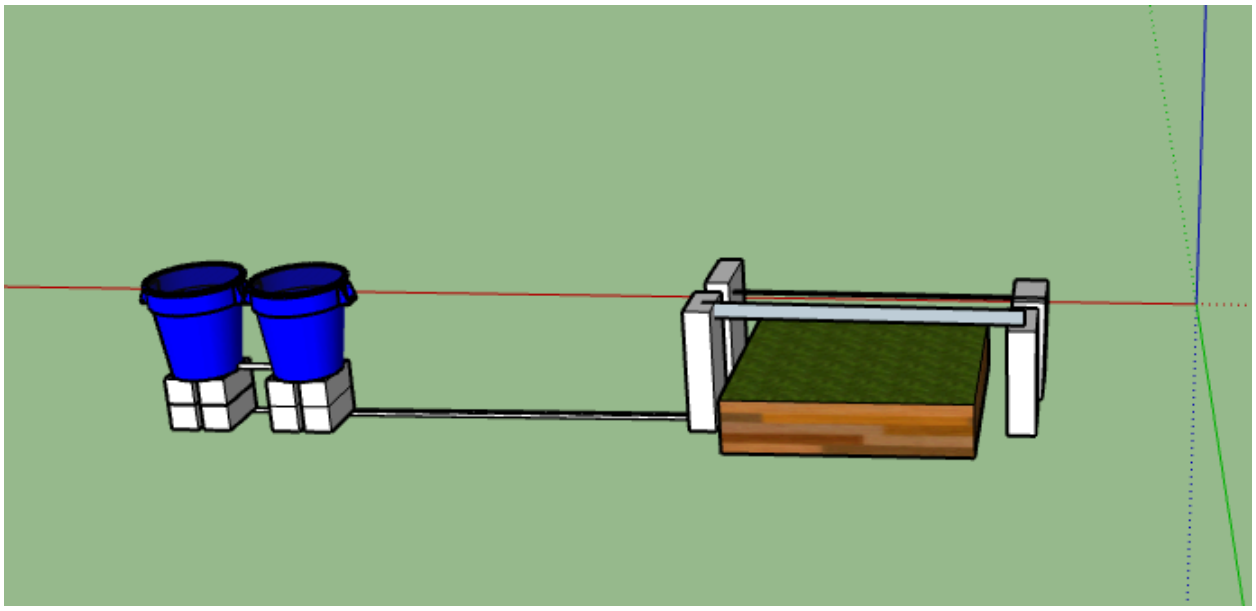
water shoots/drips out of the holes watering the plants.



/////

Aidan's Design

Date: 2/26
Description: I made a drip irrigation system. The water comes from the rain barrel then goes through the pipe. It goes up the support then out to the pipes that will disperse the water out over the garden
Drawing that includes distinguishing information such as scale, dimensions, and materials:

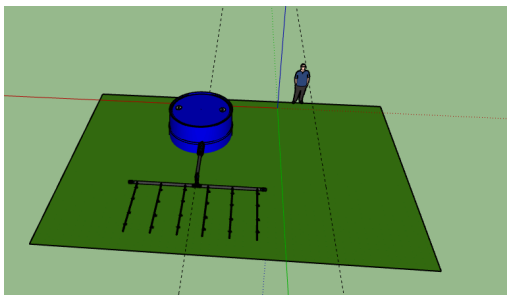


Name: _____ **Date:** _____ **Class:** _____

Davis Pelton

Date: 2/26

Description:

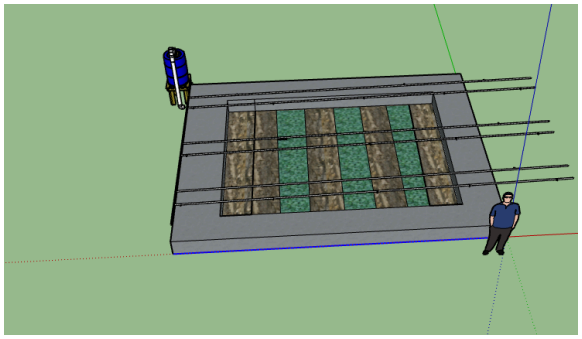


My design is a drip irrigation system that is gravity fed. It is on a timer and only turns on for 10 min every 12 hours.

Lauren's Design

Date: 2/24/25

Description: My design is 12x12 irrigation system with a rain barrel which sends water to the pipes lined above the crops to produce water.



Name: _____ **Date:** _____ **Class:** _____

Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Name: _____ **Date:** _____ **Class:** _____

	weight	Sammi	lauren	Davis	Aidan	Peter
Portable	2	1	1	0	1	0
Automatic	2	1	1	1	1	1
Holds water	2	1	1	1	1	1
Lasts 10 days	3	1	1	1	1	1
12x12	1	1	1	1	1	1
score	10	5	5	4	5	4

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1: peters design

Date:
Sketch:
Manufacturing Methods:
Step 1:build the structure
Step 2: work on making the water transport/ water pressure
Step 3: adjust the water timer
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Water timer	$12.98	1	$12.98
Wood	$2.25	4	$9.00
Pvc pipe	$11.00	1	$11.00

Name: _____ **Date:** _____ **Class:** _____

Total Cost	$32.98		

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.